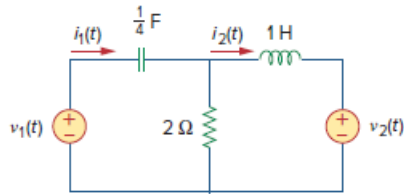


Problem

Develop the state equations for the circuit shown in Fig. 16.104.

Figure 16.104:



Step-by-step solution

Step 1 of 7

Refer to figure 16.72 for the circuit in the textbook.

Redraw the circuit as shown in figure 1.

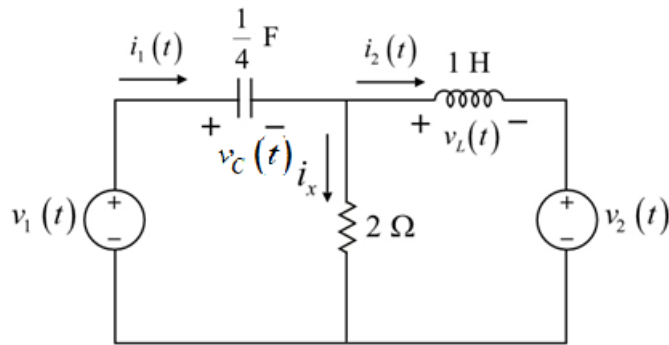


Figure 1

Step 2 of 7

Consider the inductor current, $i_L(t)$ and capacitor voltage, $v_c(t)$ as the state variables.

Thus,

$$i_2(t) = i_L(t) \dots\dots (1)$$

Calculate the current, $i_1(t)$ through the capacitor.

$$i_1(t) = C \frac{dv_c(t)}{dt}$$

$$i_1(t) = \frac{1}{4} \frac{dv_c(t)}{dt} \dots\dots (2)$$

Step 3 of 7

Calculate the current through the resistor, $i(t)$.

$$\begin{aligned} i(t) &= i_1(t) - i_2(t) \\ &= \frac{1}{4} v_c(t) - i_L(t) \end{aligned}$$

Apply Kirchhoff's voltage law in the left loop.

$$v_1(t) - v_c(t) - 2\left(\frac{1}{4} v_c(t) - i_L(t)\right) = 0$$

$$v_1(t) - v_c(t) - \frac{1}{2} v_c(t) + 2i_L(t) = 0$$

$$2v_1(t) - 2v_c(t) - v_c(t) + 4i_L(t) = 0$$

Rewrite the above equation as:

$$v_c(t) = 4i_L(t) - 2v_c(t) + 2v_1(t) \dots\dots (3)$$

Step 4 of 7

Apply Kirchhoff's voltage law in right loop of Figure 1.

$$-2i(t) + L \frac{di_L}{dt} + v_2(t) = 0$$

$$-2\left(\frac{1}{4} v_c(t) - i_L(t)\right) + (1) \dot{i}_L(t) + v_2(t) = 0$$

$$-\frac{1}{2} v_c(t) + 2i_L(t) + \dot{i}_L(t) + v_2(t) = 0$$

Substitute $4i_L(t) - 2v_c(t) + 2v_1(t)$ for $v_c(t)$.

$$-\frac{1}{2}(4i_L(t) - 2v_c(t) + 2v_1(t)) + 2i_L(t) + \dot{i}_L(t) + v_2(t) = 0$$

$$-2i_L(t) + v_c(t) - v_1(t) + 2i_L(t) + \dot{i}_L(t) + v_2(t) = 0$$

$$\dot{i}_L(t) = -v_c(t) + v_1(t) - v_2(t) \dots\dots (3)$$

Step 5 of 7

Substitute $4i_L(t) - 2v_c(t) + 2v_1(t)$ for $v_c(t)$ in (1).

$$i_1(t) = \frac{4i_L(t) - 2v_c(t) + 2v_1(t)}{4}$$

$$i_1(t) = i_L(t) - \frac{1}{2} v_c(t) + \frac{1}{2} v_1(t) \dots\dots (4)$$

Step 6 of 7

Equations (1), (2), (3) and (4) form the set of state equations.

Equation (1) and (4) are the output equations, expressing currents, $i_1(t)$ and $i_2(t)$ as the in terms of state variables.

Step 7 of 7

Rewrite equations (2) and (3) in matrix form.

$$\begin{bmatrix} \dot{i}_L(t) \\ \dot{v}_C(t) \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} i_L(t) \\ v_C(t) \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} \dots\dots (5)$$

Rewrite equations (1) and (4) in matrix form as:

$$\begin{bmatrix} i_1(t) \\ i_2(t) \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} i_L(t) \\ v_C(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{2} & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} \dots\dots (6)$$

Therefore, the state equations are described by equations (5) and (6) as:

$$\boxed{\begin{bmatrix} \dot{i}_L(t) \\ \dot{v}_C(t) \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} i_L(t) \\ v_C(t) \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}} \text{ and}$$

$$\boxed{\begin{bmatrix} i_1(t) \\ i_2(t) \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} i_L(t) \\ v_C(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{2} & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}}.$$